

Name: Srinjoy Chakraborty

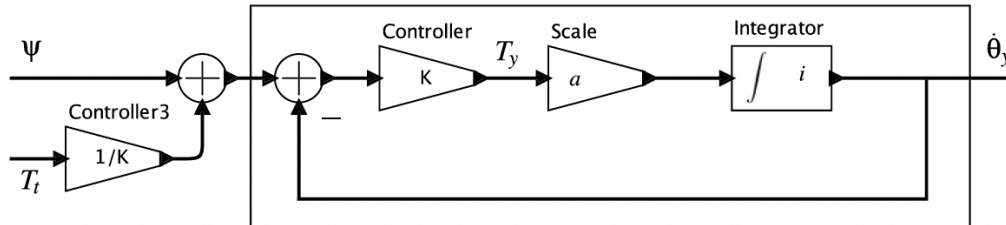
Indian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical SystemsRoll No: 251040638
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Homework Assignment 1

Deadline: September 11, 2026

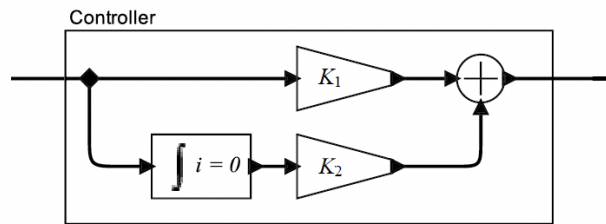
Total: 60 marks**Problem 1.** (10 points)

(a) Construct a Simulink model of the helicopter control system shown in the figure below.



Choose some reasonable parameters and plot the actual angular velocity as a function of time, assuming that the desired angular velocity is zero, $\psi(t) = 0$, and that the top-rotor torque is non-zero, $T_t(t) = bu(t)$. Give your plot for three different values of K and discuss how the behavior varies with K .

(b) Modify the model of part (a) to replace the Controller (the simple scale-by- K actor) with the alternative controller shown in the figure below.



This alternative controller is called a proportional-integrator (PI) controller. It has two parameters K_1 and K_2 . Experiment with the values of these parameters, give some plots of the behavior with the same inputs as in part (a), and discuss the behavior of this controller in contrast to the one of part (a).

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Name:

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Homework Assignment 1

Deadline: September 11, 2026

Problem 2. (10 points)

The states of the linearized model of a vehicle steering system represent the lateral deviation of the vehicle from the x -axis and the angle between the vehicle axis and the x -axis. The output of the linearized model is only the first state. Construct a Simulink model for the vehicle steering system with its controller that includes an observer. The dynamics are available in Example 6.4 and Example 7.3 in [AM09]. Apply a sinusoidal signal as the reference trajectory that specifies the desired deviation of the vehicle from the x -axis with time. Plot the output (lateral deviation of the vehicle from the x -axis) with time.

[AM09] K. J. Astrom and R. M. Murray. Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press, 2009.

http://www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_22Feb09.pdf .

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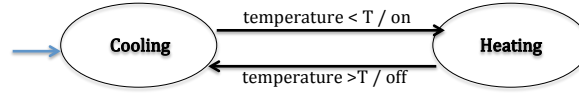
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Homework Assignment 1

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Problem 3. (10 points)

Consider the following model for a thermostat system.



The thermostat has been designed to maintain the temperature of a room at $T^{\circ}\text{C}$. The model has two states: *cooling* and *heating*. When the system is in the *cooling* state and the temperature of the room goes below $T^{\circ}\text{C}$, the system generates a signal to switch on a heater and moves to the *heating* state. When the temperature of the room goes over $T^{\circ}\text{C}$, the system generates a signal to switch off the heater and moves to the *cooling* state.

- Represent the system as an actor that takes the current temperature as input and produces a signal to control the heater. The actor uses the set point T as a parameter.
- Identify a design problem in the model.
- Provide two different remedies to address the problem.
- Compare your proposed two solutions in terms of ease of implementation and guarantees on the system behavior.

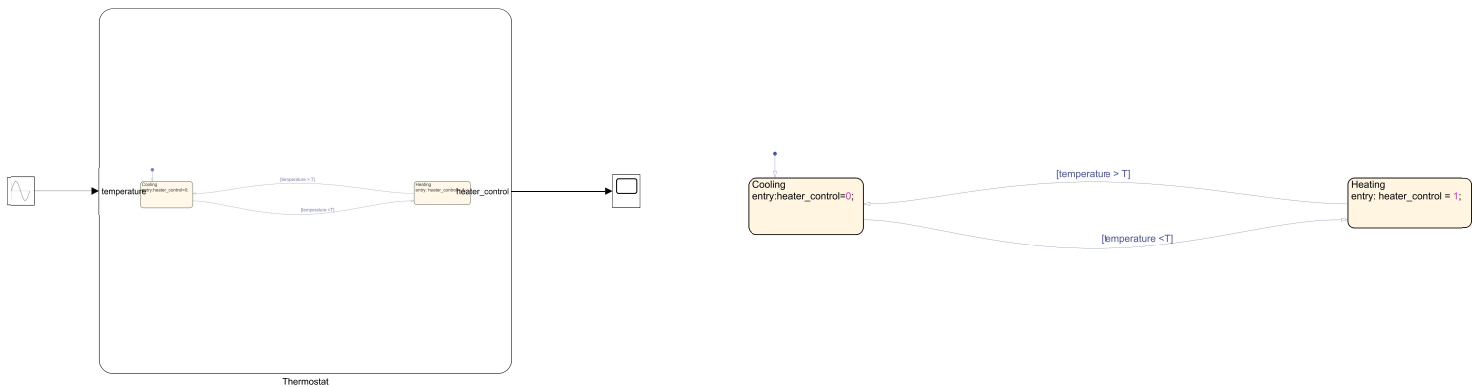
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The thermostat system is represented as a Stateflow chart actor in Simulink. The actor takes `temperature` as a continuous input signal and produces `heater_control` as a discrete output signal.

Initializations: The set point parameter T is defined in the MATLAB workspace (e.g., using `T = Simulink.Parameter; T.Value = 24;`). The variable T is assigned the Scope of “Parameter” in the Stateflow Symbols pane, allowing the chart to read it during execution.

Connections: On the Simulink canvas, a Sine Wave generator is connected to the `temperature` input port to simulate room temperature fluctuations. The `heater_control` output port is wired to a Scope block alongside the temperature input to visualize the state transitions over time.



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e.g. CSE

Homework Assignment 1

Deadline: September 11, 2026

(b) Design Problem

The primary design flaw in this model is **chattering** (or Zeno behavior). Because there is only a single absolute threshold (T), any minor sensor noise or rapid micro-fluctuations around $T^{\circ}\text{C}$ will cause the system to transition back and forth continuously. This high-frequency toggling can cause severe physical damage to the heater's mechanical relays and drastically reduce its lifespan.

(c) Proposed Remedies

To resolve the chattering issue, the following two remedies can be implemented:

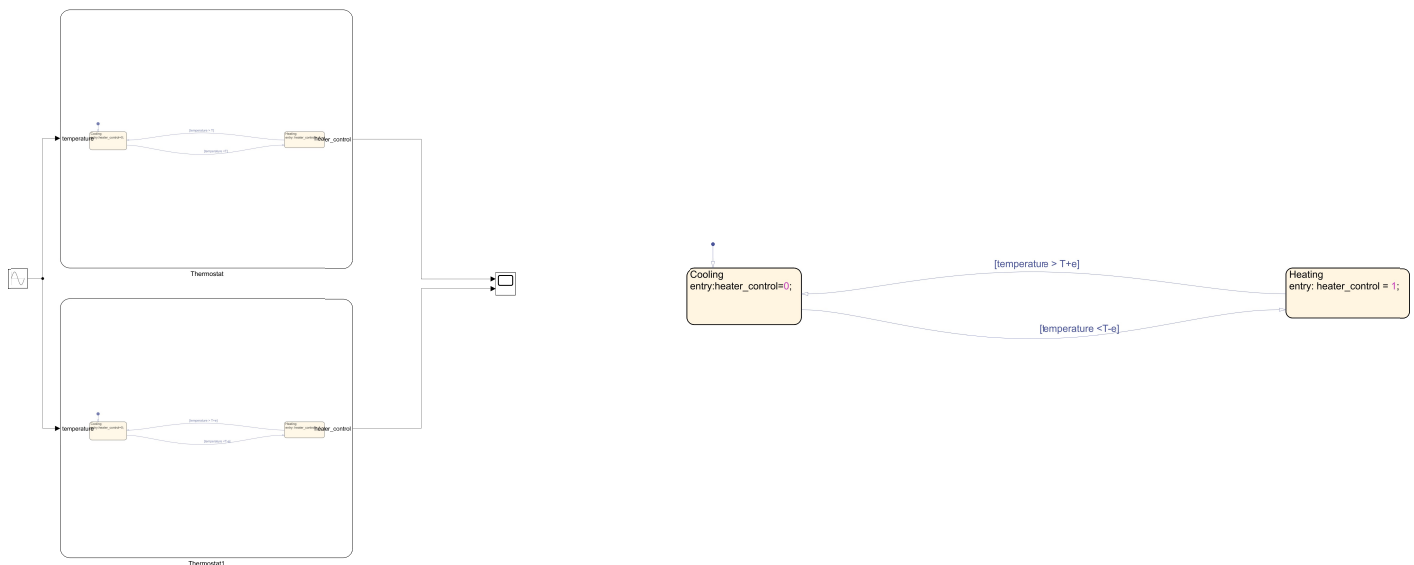
1. **Hysteresis (Spatial Thresholding):** Introduce an error margin or tolerance, e . The system is modified to turn the heater on at $T - e$ and off at $T + e$. This creates a "deadband" that safely absorbs small temperature noise.
2. **Temporal Delay (Minimum Dwell Time):** Enforce a strict duration timer using temporal logic (e.g., `after(5, sec)`). Once the heater switches state, it is forced to remain in that state for a fixed minimum duration before it is allowed to transition again, regardless of temperature fluctuations.

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Deadline: September 11, 2026**(d) Comparison of Solutions**

Ease of Implementation: Hysteresis is very simple to implement, as it only requires altering the algebraic logic on the transition arrows (e.g., changing `[temperature < T]` to `[temperature < T-e]`). The temporal delay is slightly more complex, requiring absolute-time temporal operators and potentially hierarchical sub-states to manage the timers cleanly.

Guarantees on System Behavior: Hysteresis guarantees stability against noise bounded within the $2e^{\circ}\text{C}$ band, but if the room temperature itself swings violently, rapid switching could still theoretically occur. The temporal delay, however, provides a *strict, absolute guarantee* on the physical hardware's maximum switching frequency, ensuring the heater is never toggled faster than the defined dwell time under any circumstances.



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Problem 4. (10 points)

Construct a timed automation model of a thermostat where the heater will be switched on when the room temperature is less than 22C and switched off when the temperature is above 26C. We also have the following additional requirement. Before switching on or off, it will generate a “beep” sound every second for 5 seconds as a warning. That means once the system senses that the temperature is less than 22C, it will generate the warning first, and then the heater will be switched on. Similarly, a warning sound will be issued before switching off the heater after the temperature reaches 26C.

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Deadline: September 11, 2026

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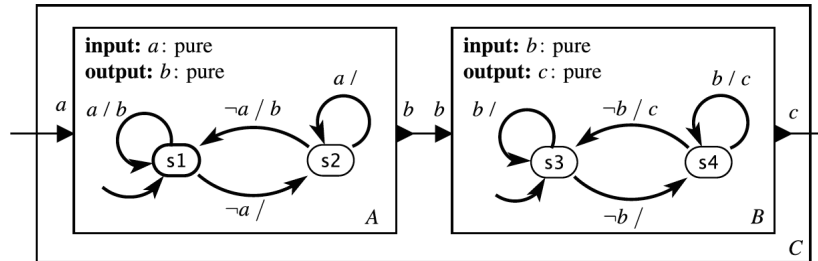
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Consider the following synchronous composition of two state machines A and B.



Construct a single state machine C representing the composition. Which states of the composition are unreachable?

Name:

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Roll No:
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Deadline: September 11, 2026

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Homework Assignment 1
Deadline: September 11, 2026

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Homework Assignment 1
Deadline: September 11, 2026

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Homework Assignment 1

Deadline: September 11, 2026

Problem 6. (10 points)

Consider the following SDF model:



The numbers adjacent to the ports indicate the number of tokens produced or consumed by the actor when it fires. Answer the following questions about this model.

(a) Let q_A , q_B , and q_C denote the number of firings of actors A , B , and C , respectively. Write down the balance equations and find the least positive integer solution.

(b) Find a schedule for an unbounded execution that minimizes the buffer sizes on the two communication channels. What is the resulting size of the buffers?

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Homework Assignment 1
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